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Comprehensive assessment of coolants used single-phase immersion cooling data center

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ABSTRACT

Currently, the single-phase immersion cooling (SPIC) data center has received much attentions due to its high cooling efficiency, capability for high-density deployment, and reduced maintenance costs. Central to the performance of SPIC system is the choice of coolant, yet a comprehensive assessment of various coolants is still lacking. This study addresses this gap by comparing the performance of six different coolants from thermal performance, energy efficiency, economic evaluation, environmental impact, and regional variations. The findings indicate that fluorocarbon coolants generally offer superior thermal performance. Specifically, the figure of merit (FOM) for SF10 is between 17.2 % and 50.6 % higher than other coolants at a coolant temperature of 40 °C. In contrast, hydrocarbon coolants excel in energy efficiency, cost-effectiveness, and environmental impact. For instance, BINGYI 797 reduces energy consumption by 14–54 %, cuts coolant costs by 5.7–94.6 %, and decreases CO_2 emissions by 13.9–97.4 % compared to other coolants. The study also notes regional variations in coolant performance. Depending on the coolant used, energy consumption can vary from 18.9 % to 57.3 %, power usage effectiveness (PUE) could range from 1.039 to 1.150, life cycle cost (LCC) could differ by 13.7–61.5 %, and CO_2 emissions can fluctuate between 1.8 % and 81.5 %. Based on the evaluated six coolants, SF10 and BINGYI 797 are current the most promising candidates for fluorocarbon and hydrocarbon coolants, respectively.

Nomenclature

C_{CT}	Coolant costs for the entire system, CNY	RFD	CO_2 emission of coolant disposal, $kgCO_2e/kg$
C_E	Purchasing cost of SPIC equipment, CNY	RFM	CO_2 emission of coolant manufacturing, $kgCO_2e/kg$
C_{en}	Energy consumption cost, CNY	S_s	Solid source term
$C_{current}$	Current heat capacity flow rate, kJ/°C	T_{out}	The temperature of the fluid exiting the device, °C
C_h	Heat capacity flow at high temperature side of plate heat exchanger, kJ/°C	T	Temperature, °C
		T_A	Average GPU temperature, °C
		T_c	Lifetime of the SPIC system, years
C_{min}	Minimum heat capacity flow in the low and high temperature sides of plate heat exchangers, kJ/°C	T_s	Solid temperature, °C
		T_{in}	Temperature of the fluid flowing into the device, °C
c_{elec}	Electricity price, CNY/kWh	T_{hin}	Inlet temperature of high temperature side of plate heat exchanger, °C
c_{crb}	Carbon price, CNY/ton		
C_{om}	Operation and maintenance cost, CNY	T_{hout}	Outlet temperature of high temperature side of plate heat exchanger, °C
c_p	Constant pressure specific heat capacity, J/(kg·°C)	T_{cin}	Inlet temperature of low temperature side of plate heat exchanger, °C
C_r	Replenishing cost of the coolant, CNY	u	Fluid velocity along the x-axis direction, m/s
E_{direct}	CO_2 emissions from direct, $kgCO_2$	v	Fluid velocity along the y-axis direction, m/s
$E_{indirect}$	CO_2 emissions from indirect, $kgCO_2$	w	Fluid velocity along the z-axis direction, m/s

(continued on next page)

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(*continued*)

E_{leak}	CO_2 emissions from the leakage of coolant, $kgCO_2$
E_{accid}	CO_2 emissions from accident, $kgCO_2$
E_{energy}	CO_2 emissions from energy consumption, $kgCO_2$
E_{rec}	CO_2 emissions from the recycle of coolant, $kgCO_2$
E_{man}	CO_2 emissions from the manufacture of coolant, $kgCO_2$
E_{deal}	CO_2 emissions from the deal of coolant, $kgCO_2$
$f_{motorloss}$	The rate at which the pump motor transfers heat to the fluid
g	Gravitational acceleration, m/s^2
$C_{\varepsilon 1}$	Constant
$C_{\varepsilon 2}$	Constant
$C_{\varepsilon 3}$	Constant
G_b	Turbulent kinetic energy generated by buoyancy
G_κ	Turbulent kinetic energy at the mean velocity gradient
h	Heat-transfer coefficient, $W/(m^2 \cdot ^\circ C)$
in	Discount rate
L	Coolant amount, m^3
m	CPU number
m_{fluid}	Mass flow of fluid, kg/h
Mo	Mouromtseff number
m_r	Rated mass flow of pump, kg/h
n	GPU number
N	Pump shaft power, W
Nu	Nusselt number
P	Pressure, Pa
Pr	Prandtl number
P_{CPU}	CPU power consumption, W
P_{GPU}	GPU power consumption, W
P_p	Pump power, W
PW	Present Value, CNY
$Q_{cooling}$	Annual energy consumption of cooling equipment, MW·h
Q_{IT}	Annual energy consumption of IT equipment, MW·h
$q_{current}$	Current heat exchange volume, kJ/h
q_w	Heat transferred from the pump motor to the fluid, kJ/h
q_H	Heat transfer of plate heat exchanger, kJ/h
q_v	Volume flow rate, m^3/h
$W_{cooling}$	Total power consumption of cooling equipment, W
W_{cp}	Power consumption of coolant pump, W
W_{cwp}	Power consumption of cooling water pump, W
W_{dry}	Power consumption of dry cooler, W
x	Coordinate axis
y	Coordinate axis
z	Coordinate axis
Greek symbol	
α	Energy efficiency of plate heat exchangers
β	expansion coefficient
λ	Thermal conductivity, $W/(m \cdot ^\circ C)$
λ_f	Fluid thermal conductivity, $W/(m \cdot ^\circ C)$
λ_s	Solid thermal conductivity, $W/(m \cdot ^\circ C)$
σ_ε	Constant
σ_κ	Constant
ε	Dissipation rate of turbulent pulsation kinetic energy
κ	Turbulent pulsation kinetic energy
ρ	Density, kg/m^3
η_{pump}	Pump efficiency
μ	Dynamic viscosity, mPa·s
γ_p	Pump control signal
τ	Operating time, h
Abbreviations	
Adp. GWP	GWP of atmospheric degradation production of coolant, $kgCO_{2e}/kg$
ALR	Annual leakage rate
CFD	Computational fluid dynamics
CRF	Capital recovery factor
CPU	Central processing unit
EOL	Emissions generated at the end of coolant life
FCI	Fixed capital investment, CNY
FOM	Figure of merit
FVM	Finite volume method
GWP	Global warming potential
GPU	Graphics processing unit
LCC	Life cycle cost, CNY
LCCP	Life cycle climate performance
ODP	Ozone depletion potential
PEC	Purchasing equipment cost, CNY
PID	Proportion Integral Differential
PUE	Power Usage Effectiveness
RAM	Random-access memory
SPIC	Single-phase immersion cooling
TCO	Total Cost of Ownership, CNY

1. Introduction

With the advent of cloud computing and advancements in artificial intelligence, the global demand for computational resources is escalating, placing significant pressure on data centers (DCs), which serve as the pivotal hubs for information processing and storage, to manage their substantial energy consumption [1,2]. The cooling systems, as primary consumers of this energy, are crucial in shaping the overall energy efficiency of DCs [3]. To enhance energy efficiency, curtail carbon emissions, and accommodate the burgeoning demand for computing capabilities, the deployment of liquid cooling technologies is gaining increasing attention. Among these, the single-phase immersion cooling (SPIC) system has emerged as a focal point in contemporary cooling technologies due to its superior cooling efficiency [4], capability for high-density deployment [5], and reduced maintenance expenses [6,7]. In this approach, servers are entirely submerged in a dielectric coolant, facilitating heat dissipation with markedly improved efficiency and energy utilization compared to conventional air-based cooling methods, thereby presenting extensive potential for application [8,9].

Since the servers are fully submerged in the coolant, the coolant's thermophysical properties critically influence system performance, such as heat dissipation efficiency, thermal resistance, operational stability, and long-term reliability. Currently, two prominent categories of coolants are in use: fluorocarbon coolants and hydrocarbon coolants [10]. Fluorocarbon coolants, such as perfluorinated compounds, have gained attention due to their excellent dielectric properties, high thermal conductivity, and chemical stability [11,12]. Despite these advantages, concerns have been raised about their potential environmental impact and toxicity [13,14]. In contrast, hydrocarbon coolants, such as polyalphaolefins (PAOs) and polyalkylene glycols (PAGs), are considered as viable alternatives [15]. These coolants offer favorable thermal and lubricating properties and are generally more environmentally friendly than their fluorinated counterparts [16]. However, a systematic, quantitative evaluation of the existed coolants on comprehensive performance and lifecycle environmental impact remains lacking in current literature.

Currently, the thermal performance evaluation of coolants employed in SPIC system has attracted considerable research interest, which includes evaluating the thermophysical properties of coolants, comparing the performance of different coolants, and optimizing the compatibility between servers and coolants. Modification of key thermophysical parameters, such as particularly dynamic viscosity, thermal conductivity, and specific heat capacity, could significantly alter heat transfer efficiency. For instance, Hnayno et al. [17] studied the impact of dynamic viscosity and found that increasing the viscosity of a coolant from 4.6 mPa s to 9.8 mPa·s resulted in a coolant temperature rise of about 6 %. Zhou et al. [18] demonstrated that increasing thermal conductivity and specific heat capacity can lower the temperature of the heat source by 3.4 °C–4.8 °C. Similarly, Wen et al. [19] enhanced the thermal conductivity of FC-40 by incorporating Al nanoparticles, achieving a 6.5 % reduction in maximum GPU temperature. Luo et al. [20] conducted comparable research, showing that the addition of SiC nanoparticles to mineral oil could enhance the heat transfer coefficient by 11.7 %.

A systematic comparison of coolant thermal performance is critical for identifying optimal candidates for immersion cooling applications. For example, Wang et al. [21] compared Novec 7100 and transformer oil in a buoyancy-driven SPIC system. Despite transformer oil having a thermal conductivity five times greater than that of Novec 7100, its high viscosity, 90 times higher than that of Novec 7100, diminished the thermal efficiency of natural convection, which resulted in the SPIC system using transformer oil maintaining a temperature 3 °C higher than when Novec 7100 is used. Choi et al. [22] assessed temperature uniformity in SPIC system with Novec 7500 and SF10 coolants, finding that SF10 achieved a temperature difference of only 5 °C, indicating superior temperature uniformity compared to Novec 7500. Chen et al. [23] evaluated three electronic fluorinated liquids, concluding that using a coolant with the lowest dynamic viscosity increased the average Nusselt number (*Nu*) of the SPIC system by 57.3 % and reduced the pressure drop by 59 %. Wang et al. [4] demonstrated that reducing dynamic viscosity is crucial, as a decrease in the average *Nu* by 24.4 % and an increase in flow resistance loss by 7.4 times were observed with less optimal viscosity choices. However, the comprehensive comparison about the existing coolants is still lacking.

Optimal cooling performance also requires careful alignment of coolant thermophysical properties with server architecture characteristics. Li et al. [24] investigated the impact of various fin parameters, such as thermal conductivity, fin height, fin spacing, and baffle height ratio, on cooling efficiency. They discovered that an improved fin radiator with a fin height of 25 mm, fin spacing of 5 mm, and thermal conductivity of 300 W/(m·°C) significantly reduced the maximum temperature by 12.5 °C. Muneeshwaran et al. [25] examined the flow characteristics of FC-40 coolant in servers with different structural designs. Their findings indicated that in T-shaped servers, the thermal resistance and shell temperature of FC-40 decreased by 12.6 % and 0.5–2.8 °C, respectively, when compared to Z-shaped servers. Shrigondekar et al. [26] analyzed the heat transfer behavior of coolants (FC-40 and PAO-6) in SPIC system with varying fin spacing. They found that, for PAO-6, a fin spacing of 2.2 mm yielded better performance than a denser spacing of 1.2 mm, while for FC-40, the denser spacing performed better. Taddeo and Barestrand et al. [27,28] investigated the thermal performance of a SPIC system under static and dynamic workloads with oil as the coolant. Their study established a correlation for server energy use across a range of system temperatures.

The existing literatures demonstrate extensive investigations into the thermal performance of coolants. To enhance the practical applicability of these findings, the analysis should be expanded to the energy efficiency, economic evaluation, and environmental impact assessments, based on both existing coolant classes. Therefore, this study aims to conduct a comprehensive evaluation of both fluorocarbon and hydrocarbon coolants used in SPIC system. To achieve the balanced comparison, the study begins with a detailed analysis of the thermal performance of these coolants by using the validated numerical model. This will be followed by assessments of energy consumption, which is crucial for operational efficiency, and an economic evaluation to determine the cost-effectiveness of each type of coolant. Additionally, the environmental impact of using different coolants is also analyzed. The conclusions drawn from this study would be valuable for selecting the most suitable coolants for SPIC system, providing a holistic view that goes beyond thermal performance alone.

2. Description of data center and methodology

2.1. Description of studied data center

In the examined SPIC system, the assembly comprises a liquid cooling tank, a plate heat exchanger, a dry cooler, a coolant circulation pump, and a cooling water pump, as illustrated in Fig. 1(a). During operation, the coolant pump facilitates the circulation of coolant through the liquid cooling tank, effectively dissipating the heat generated by servers. Subsequently, the thermally elevated coolant is directed to the plate heat exchanger, where it transfers its heat to the cooling water, thereby completing the cycle of coolant circulation. Concurrently, the cooling water, a 20 % ethylene glycol aqueous solution, absorbs the heat emitted from the coolant and discharges it within the dry cooler. The heat transfer efficiency between cooling water and ambient air in dry cooler exhibits strong temperature dependence [29,30]. The plate heat exchanger in the SPIC-DC system represents a crucial component, facilitating the heat exchange between the coolant and the cooling water. Additionally, the liquid cooling tank in this study accommodates 20 servers with a total power capacity of 124 kW. Each server, drawing power of 6.2 kW, predominantly derives its power from two central processing units (CPUs, Intel® Xeon® Platinum 8473C, 350 W) and ten graphics processing units (GPUs, NVIDIA GeForce RTX 4090, 550W), as depicted in Fig. 1(b).

2.2. Coolants information

The coolants presently employed within SPIC system are categorized into fluorocarbon and hydrocarbon varieties. The fluorocarbon coolant can be future devided into perfluoroamines and hydrofluoroethers, where FC-40, Novec 7500, and SF10 are widely used in the open literatures [22,25]. For the hydrocarbon coolant, the synthetic hydrocarbon oil has received more attention, where VP-3, EC-110, and BINGYI 797 are employed in this study [31–33]. The specific thermophysical properties of each coolant are delineated in Table 1.

Based on the studied SPIC system and the collected coolants, a thorough comparison of the various coolants could be conducted, which includes thermal performance, energy consumption, economic evaluation, and environmental impact.

2.3. Methodology

In this study, both Computational Fluid Dynamics (CFD) and TRNSYS models are employed to provide a comprehensive analysis of different coolants. The CFD model is primarily utilized to simulate the thermal performance of different coolants within servers. The CFD simulation results, especially for the relationship between average GPU temperature and coolant flowrate and temperature, serve as primary inputs for the TRNSYS model. Then, the system-level performance by using different coolants is analyzed, including the energy consumption, economic evaluation, and environmental impact.

2.3.1. CFD model for the server

The commercial software FloEFD is employed to simulate the heat and mass transfer within the server. Drawing from the aforementioned physical server model, this study constructs a 3D steady-state representation. The governing equations are delineated as follows:

Continuity equation can be found in the following:

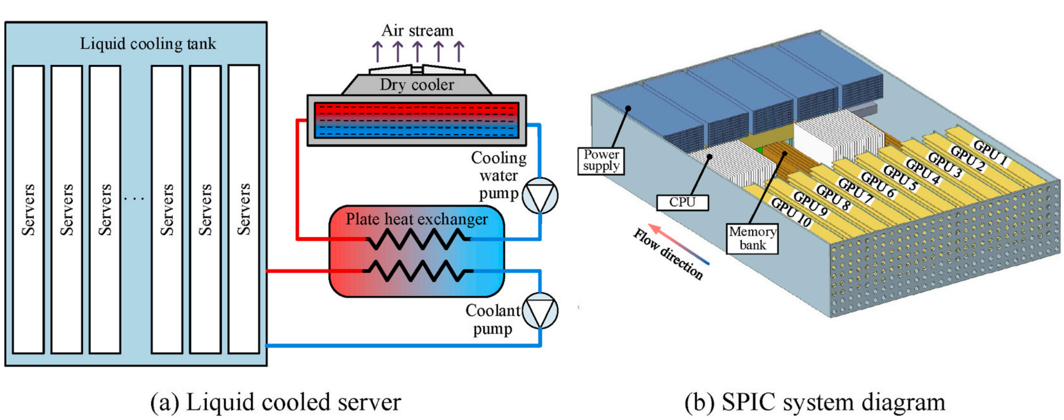


Fig. 1. Schematic of the single-phase immersion cooling data center.

Table 1
Thermophysical properties of coolants [34–38].

Item		Fluorocarbon coolant			Hydrocarbon coolant		
		FC-40	Novec 7500	SF10	BINGYI 797	VP-3	EC-110
Pourpoint, °C		−57	−100	−90	−65	2.4	−60
Boiling point, °C		165	128	110	300	243	/
Density, kg/m^3	20 °C	1866	1624	1589	794	933	820[a]
	40 °C	1823	1582	1558	786	923	/
	60 °C	1779	1541	1524	775	911	/
	80 °C	1736	1499	1487	762	899	/
Viscosity, mPa·s	20 °C	4.66	1.36	1.26	11.72	2.95	16.54
	40 °C	2.73	0.98	0.87	6.05	1.87	5.88
	60 °C	1.78	0.74	0.63	3.63	1.33	3.55
	80 °C	1.22	0.58	0.48	2.40	1.02	2.44
Heat capacity, J/(kg·°C)	20 °C	1045	1121	1225	1965	1604	2136
	40 °C	1076	1151	1252	2051	1698	2212
	60 °C	1107	1181	1283	2154	1789	2288
	80 °C	1138	1211	1314	2253	1876	2363
Thermal conductivity, W/(m·°C)	20 °C	0.066	0.066	0.079	0.336	0.117	0.139
	40 °C	0.064	0.062	0.076	0.360	0.115	0.137
	60 °C	0.063	0.060	0.073	0.380	0.112	0.136
	80 °C	0.061	0.057	0.070	0.423	0.110	0.135
ODP		0	0	0	0	0	0
GWP		5000	100	2.5	0	0	0
Price, CNY/kg		700	1000	530	110	100	166

[a] The coolant density under different temperature can be calculated by thermal expansion coefficient (0.00067 vol/°C).

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0 \tag{1}$$

Momentum equation is described as follows:

$$\begin{cases} \rho\left(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)=-\frac{\partial P}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right) \\ \rho\left(u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\right)=-\frac{\partial P}{\partial y}+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right) \\ \rho\left(u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right)=-\frac{\partial P}{\partial z}+\mu\left(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\right)-\rho g_z \end{cases} \tag{2}$$

Energy equations can be found as follows:

$$\begin{cases} \rho c_p\left(u\frac{\partial T_f}{\partial x}+v\frac{\partial T_f}{\partial y}+w\frac{\partial T_f}{\partial z}\right)=\lambda_f\left(\frac{\partial^2 T_f}{\partial x^2}+\frac{\partial^2 T_f}{\partial y^2}+\frac{\partial^2 T_f}{\partial z^2}\right) \quad \textit{Fluid} \\ \lambda_s\left(\frac{\partial^2 T_s}{\partial x^2}+\frac{\partial^2 T_s}{\partial y^2}+\frac{\partial^2 T_s}{\partial z^2}\right)+S_s=0 \quad \textit{Solid} \end{cases} \tag{3}$$

where u, v, and w are the velocities in the x, y, and z directions, m/s; ρ is the density of the coolant, kg/m^3; g is the acceleration of gravity, m/s^2; μ is the dynamic viscosity, Pa·s; c_p is the specific heat capacity of the coolant, J/(kg·°C); λ_f and λ_s are the thermal conductivity of coolant and solid, W/(m·°C), respectively. T_f and T_s are the temperature of coolant and solid respectively, °C.

Based on theoretical research [39], the standard κ-ε model is employed, which can be described as follows:

$$\begin{cases} \frac{\partial}{\partial x_i}(\rho u_i \kappa)=\frac{\partial}{\partial x_j}\left[\left(\mu+\frac{\mu_t}{\sigma_\kappa}\right)\frac{\partial \kappa}{\partial x_j}\right]+G_\kappa+G_b-\rho\varepsilon \\ \frac{\partial}{\partial x_i}(\rho u_i \varepsilon)=\frac{\partial}{\partial x_j}\left[\left(\mu+\frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right]+C_{\varepsilon 1}\frac{\varepsilon}{\kappa}(G_\kappa+C_{\varepsilon 3}G_b)-C_{\varepsilon 2}\frac{\rho\varepsilon^2}{\kappa} \end{cases} \tag{4}$$

where i and j represents the x-, y-, or z-directions, respectively; κ is turbulent pulsation kinetic energy; ε is the dissipation rate of turbulent pulsation kinetic energy; μ_t is the turbulent viscosity; G_κ is the turbulent kinetic energy at the mean velocity gradient; G_b the turbulent kinetic energy generated by buoyancy; $C_{\varepsilon 1}$ (1.44), $C_{\varepsilon 2}$ (1.92), $C_{\varepsilon 3}$ (1.0), σ_κ (1.0), and σ_ε (1.3) are constants in the standard κ-ε model. μ_t, G_κ, and G_b can be defined as following:

$$\begin{cases} \mu_t=\rho C_\mu \frac{\kappa^2}{\varepsilon} \\ G_b=\beta g_i \frac{\mu_t}{Pr_t}\frac{\partial T}{\partial x_i} \\ G_\kappa=\rho\overline{u_i u_j}\frac{\partial u_j}{\partial x_i} \end{cases} \tag{5}$$

where Pr is the Prandtl number, Pr_t is 0.85, C_μ is constant which is 0.09, β is the expansion coefficient, which can be defined as following:

$$\beta=-\frac{1}{\rho}\left(\frac{\partial\rho}{\partial T}\right)_P \tag{6}$$

During the simulation, the boundary condition can be found in Table 2.

2.3.2. TRNSYS model

The TRNSYS software functions as an extensive computational platform for transient system-level simulations, facilitates analysis of the overall performance of SPIC system. Within the simulations, the temperature fluctuations within the liquid cooling tank and the coolant flow rates are derived from the validated CFD model. In conjunction with Fig. 1, the TRNSYS model is further elucidated in Fig. 2. Comprehensive details and the mathematical formulations for each component are available in Table 3. To enable equitable performance comparison

Table 2
Detailed boundary condition for the CFD simulation.

Item	Boundary condition
Thermal resistance between heat sink and CPU and GPU	0.025 °C/W [24,40]
Thermal resistance between motherboard and CPU and GPU	2.8 °C/W [24,40]
Heat loads for CPU and GPU	350 W and 550 W
Inlet volumetric flow rate	1–7 m^3/h
Inlet temperature	40 °C
Outlet	Pressure outlet, 101,325 Pa
Server shell	Adiabatic wall

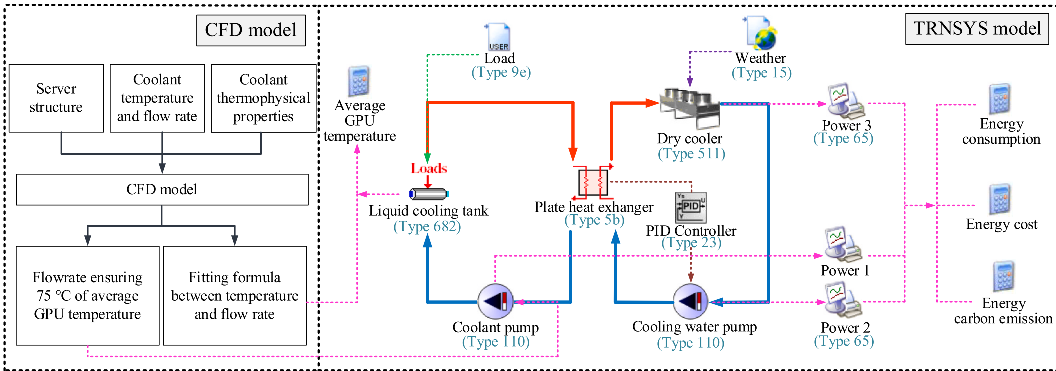


Fig. 2. TRNSYS model of SPIC-DC.

Table 3
Detailed parameters for SPIC system and the mathematical model.

Item	Parameter	Sub-model	Mathematical formulation
Heat load	Server number: 20 Power: 124 kW Size: 1850 × 1200 × 950 mm	Type 682	$T_{out} = T_{in} + \frac{W_{IT}}{m_{fluid} c_p}$
Coolant pump	Type: TD100-27G/2 Rated flowrate: 100 m^3/h Rated power: 11 kW Price: 4640 CNY	Type 110	$m_{fluid} = m_r \cdot \gamma_p$ $q_w = N(1 - \eta_{pump}) + (P_p - N) f_{motorloss}$ $T_{p,w,o} = T_{p,w,i} + q_w / m_{fluid}$
Cooling water pump	Type: TD100-22G/2 Rated flowrate: 80 m^3/h Rated power: 7.5 kW Price: 3260 CNY	Type 110	
Plate heat exchanger	Type: AC1000DQ Heat transfer coefficient: 1330.3 W/(m^2·°C) Heat exchanger area: 9.36 m^2 Price: 4400 CNY	Type 5b	$T_{hout} = T_{hin} - \alpha(C_{\min}/C_h)(T_{hin} - T_{cin})$ $q_H = \alpha C_{\min}(T_{hin} - T_{cin})$
Dry cooler	Type: LDDC-0204V-3R-AC800 Rated power: 10.8 kW Price: 36500 CNY	Type 511	$T_{Out} = T_{In} - \frac{q_{current}}{C_{current}}$
PID	Controlled variable: Coolant side outlet temperature in plate heat exchanger Setpoint: 40 °C Control signal: Cooling water pump signal	Type 23	$\begin{cases} e(t) = r(t) - y(t) \\ u(t) = K_P\left[e(t) + \frac{1}{T_I}\int_0^t e(t)dt + T_D\frac{de(t)}{dt}\right] \\ = K_P e(t) + K_I\int_1^t e(t)dt + K_D\frac{de(t)}{dt} \end{cases}$

across coolant alternatives, the liquid cooling tank's inlet temperature is maintained at 40 °C. Coolant flow rates are individually optimized according to each fluid's thermophysical properties while ensuring equivalent heat removal capacity. For the cooling water circulation, the outlet temperature of the dry cooler is stabilized at 30 °C by modulating fan power in response to meteorological conditions, employing a PID controller. Should there be an abnormal escalation in the ambient temperature, it becomes challenging to maintain the target temperature. In such scenarios, the PID controller compares the actual temperature with the predetermined set point and dispatches control signals to adjust the power input of the cooling pump, thereby regulating the flow of cooling water to ensure the temperature remains close to the designated value.

2.4. *Key performance indicator (KPIs)*

To assess the performance of different coolants, the perspectives of thermal performance, energy efficiency, economic evaluation, and environmental impact are considered.

2.4.1. *Thermal performance*

The average GPU temperature and the figure of merit (FOM) are utilized to assess the thermal performance of various coolants.

The average GPU temperature could directly compare the different performances of coolant, which can be obtained as follows:

$$T_A = \frac{1}{n}\sum_{i=1}^{n} T_i \quad (n = 200) \tag{7}$$

where T_A is average GPU temperature, n is GPU number, T_i is the temperature of i GPU.

Among the metrics available for evaluating FOM, options include the Nusselt number (Nu), the heat transfer coefficient (h), and the Mouromtseff number (Mo) [41]. In this investigation, the Mouromtseff number is chosen due to its widespread application [42,43] and its capacity to quantify the heat transfer rate when coolant flows over or through a particular geometry at a specified velocity [44]. It is derived from the classical Dittus-Boelter equation, as illustrated below [45]:

$$FOM = \mu^{-0.4}\rho^{0.8}{c_p}^{0.4}\lambda^{0.6} \tag{8}$$

where μ is the dynamic viscosity of the coolant, ρ is the density of the coolant, c_p is the specific heat capacity of the coolant, and λ is the thermal conductivity of the coolant.

2.4.2. *Energy efficiency*

To assess the energy efficiency, the energy consumption and the power usage effectiveness (PUE) are employed. The energy consumption means the total power consumption of cooling devices, including the coolant pump, cooling water pump, and dry cooler, which can be calculated as follows:

$$Q_{cooling} = W_{cooling} \cdot \tau = \left(W_{cp} + W_{cwp} + W_{dry}\right) \cdot \tau \tag{9}$$

where $Q_{cooling}$ is annual energy consumption of the cooling devices; W_{total} is the total power of cooling devices; W_{cp}, W_{cwp}, and W_{dry} are powers of coolant pump, cooling water pump, and dry cooler, respectively; τ is operating time which is considered as 8760 h for one year.

PUE is an indicator widely used in DC. A lower PUE means a higher energy efficiency, which can be calculated as follows [46,47]:

$$\begin{cases} PUE = \dfrac{\left(Q_{IT} + Q_{cooling}\right)}{Q_{IT}} \\ Q_{IT} = \left(n \cdot P_{GPU} + m \cdot P_{CPU}\right) \cdot \tau \end{cases} \tag{10}$$

where Q_{IT} is the energy consumption of IT equipment; P_{GPU} and P_{CPU} are the power of GPU and CPU, respectively; m is the number of CPU, which is 40.

2.4.3. Economic evaluation

For the economic evaluation, the total cost of ownership (TCO) of different coolants and the life cycle cost (LCC) of the studied SPIC system are used. TCO includes the initial cost and additional cost due to coolant leakage [48], which can be obtained as follows:

$$\boldsymbol{TCO} = C_{CT} \cdot (1 + ALR \cdot T_c) \tag{11}$$

$$C_{CT} = c \cdot L \cdot \rho \tag{12}$$

where C_{CT} is the coolant cost of the system, CNY; L is the amount of coolant in the SPIC system, 2 m^3; ρ is the density of coolant, kg/m^3; ALR is the annual leakage rate, assumed to be 5 %; T_c is the lifetime of the SPIC system, 20 years [49].

LCC is the total cost associated with a system throughout its entire lifespan. It primarily consists of two components: fixed capital investment (FCI) and future costs. Since the FCI consists of many costs that are difficult to track, according to Kanbur's research [49], FCI can be simplified as a function of purchasing equipment cost (PEC), as shown in Eqn. (14). The PEC includes the cost of coolant (C_{CT}) and the purchasing cost of SPIC equipment (C_E). Future costs typically include energy consumption cost (C_{en}), carbon emission cost (C_{crb}), operation and maintenance cost (C_{om}), and the cost of replenishing the coolant (C_r). To accurately assess future costs, it's crucial to determine their present value (PW), which allows for more precise financial evaluation over time. To do this, the capital recovery factor (CRF) is introduced, which helps to accurately reflect the actual economic impact of coolant and other associated costs over the life cycle of the system.

$$\boldsymbol{LCC} = FCI + PW_c = FCI + PW_{en} + PW_{crb} + PW_{om} + PW_r \tag{13}$$

$$FCI = 4.3 \cdot PEC = 4.3 \cdot (C_E + C_{CT}) \tag{14}$$

$$\begin{cases} PW_{en} = \dfrac{C_{en}}{CRF} = \dfrac{W_{total} \cdot \tau \cdot c_{elec}}{[in \cdot (in+1)^n]/[(in+1)^n - 1]} \\ PW_{crb} = \dfrac{C_{crb}}{CRF} = \dfrac{W_{total} \cdot \tau \cdot c_{crb} \cdot CDE}{[in \cdot (in+1)^n]/[(in+1)^n - 1]} \\ PW_{om} = \dfrac{C_{om}}{CRF} = \dfrac{1.092\% \cdot PEC}{[in \cdot (in+1)^n]/[(in+1)^n - 1]} \\ PW_r = \dfrac{C_r}{CRF} = \dfrac{5\% \cdot C_{CT}}{[in \cdot (in+1)^n]/[(in+1)^n - 1]} \end{cases} \tag{15}$$

where C_E value is 48,800 CNY (sum of equipment prices in Table 2), c_{elec} is the electricity price of 0.634 CNY/(kW·h) [50], in is the discount rate of 1.75 % [51], n is the n-the year in life cycle, c_{crb} is the price of carbon tax of 104 CNY/ton [52], CDE is the carbon dioxide emission index of 0.5688 kg/(kW·h) [53], C_{om} is defined as 1.092 % of PEC [54], C_r is 5 % of C_{CT}, C_{en}, and C_{crb} are related to the energy consumption of the system.

2.4.4. Environmental impact

To evaluate the environmental impact, the life cycle climate performance (LCCP) metric is utilized, serving as an instrument to quantify the carbon dioxide emissions associated with the SPIC system. This comprehensive measure encompasses both direct and indirect emissions, as shown in Eqn. (16). Direct emissions arise from various scenarios of coolant leakage: emissions during routine system operation (E_{leak}), emissions attributed to the leakage during maintenance or service interventions ($E_{service}$), emissions resulting from accidental leakages (E_{accid}), and emissions occurring during the recycling phase (E_{rec}). Given that all direct emissions stem from coolant leakage, they are collectively delineated in Eqn. (17). Indirect emissions, on the other hand, encompass those attributable to the system's energy consumption (E_{energy}), the emissions generated during the production of the coolant (E_{man}), and the emissions associated with the management and disposal of waste coolant (E_{deal}). These are systematically represented in Eqn. (18).

$$LCCP = E_{direct} + E_{indirect} \tag{16}$$

$$\begin{aligned} &E_{direct} = E_{leak} + E_{service} + E_{accid} + E_{rec} \\ &= L \cdot (T_c \cdot ALR + EOL) \cdot (GWP + Adp.GWP) \end{aligned} \tag{17}$$

$$\begin{aligned} &E_{inderct} = E_{energy} + E_{man} + E_{deal} \\ &= T_c \cdot Q \cdot CDE + L \cdot (1 + T_c \cdot ALR) \cdot RFM + L \cdot (1 - EOL) \cdot RFD \end{aligned} \tag{18}$$

where EOL is the coolant emission factor at the end of life, GWP is the global warming potential, $Adp.GWP$ is the CO_2 emission factor from the degradation of coolant into environment, RFM is the CO_2 emission factor from manufacturing coolant, RFD is the CO_2 emission factor from the disposal of coolant.

2.5. Model validation

2.5.1. Validation of CFD model

Given the structural complexity of the server, a structured grid is employed for meshing, with local refinement implemented in high-power component regions (e.g., GPUs, CPUs). The SIMPLE algorithm is adopted to couple velocity and pressure variables during numerical solving. This approach iteratively solves the pressure correction equation until the continuity equation is satisfied, thereby correcting the velocity field. The discretization schemes are configured as follows: The PRESTO! scheme is applied to the pressure correction equation for enhanced adaptation to high-rotation flows and rapidly varying pressure gradients; Second-order schemes are used for momentum and energy equations to balance stability and accuracy. A high-precision solver is selected to improve computational accuracy, with convergence criteria of 10^{-7} for the energy equation and 10^{-4} for other equations. CPU and GPU temperatures are monitored separately. Convergence is deemed achieved when residual targets are met and monitored parameters stabilized across iterations.

Fig. 3 presents the grid independence test for the CFD model, employing SF10, an inlet flow rate of 50 L/min, and an inlet temperature of 40 °C. The results demonstrate that the average GPU temperature decreases progressively with grid refinement until reaching 1,122,358, beyond which temperature variations remain within 0.2 °C. This grid number is consequently adopted for all subsequent coolant performance evaluations to ensure computational accuracy while maintaining reasonable solution times.

To validate the model, the experiment is conducted and one server is employed. Where all the CPUs and GPUs work at full load, ignoring the heat load fluctuation [27]. The impact of different flowrates (1–7 m^3/h) on the average GPU temperature is discussed, as shown in Fig. 4(a).

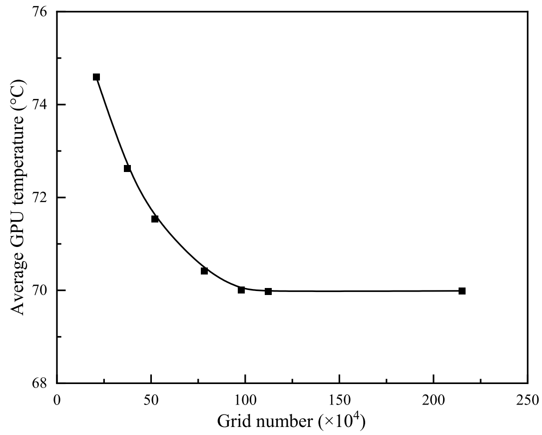


Fig. 3. Grid independence test for CFD model.

Results show that, the simulation result shows good agreement with the experimental data. The maximum temperature deviation is within 1.4 °C. Moreover, the simulation result is always lower than the experimental data, which mainly comes from the model simplification during simulation, as shown in Fig. 4 (b). Therefore, the result confirms the validity of the CFD model for comparative thermal performance analysis.

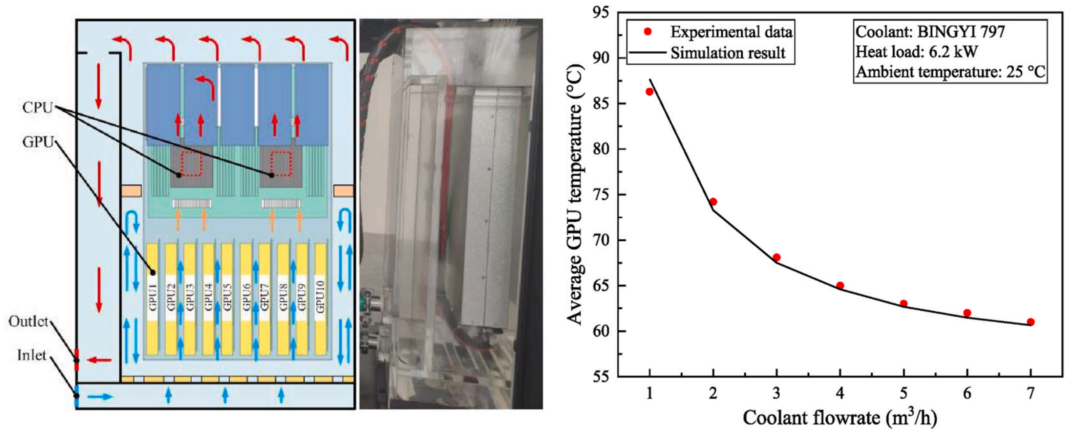
2.5.2. *Validation of TRNSYS model*

The TRNSYS model established in this study is also validated with experimental data. During the experiment, the heat load ratio of server changes from 20 % to 100 %, as show in Fig. 5(a). The result illustrates that, the simulation results of key temperatures in the system show highly agreement with the experiment data; and the maximum temperature deviation is within 0.6 °C, as shown in Fig. 5(b). Considering measurement errors, these deviations are considered as acceptable. Therefore, the model established in this study can be considered as validated and can be used to compare the different performances of coolants.

3. Results and discussion

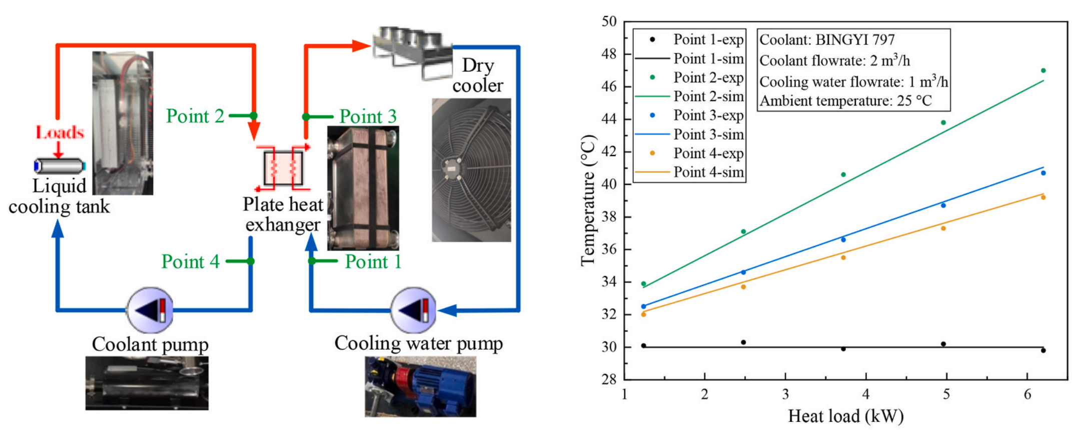
3.1. *Comparison of thermal performance*

Fig. 6 illustrates the variation in the FOM for different coolant temperatures, which comes from Table 1 and Equation (8). It is evident that as the coolant temperature increases, there is a corresponding increase in the FOM, suggesting enhanced heat transfer capacity at higher temperatures. This observed trend is predominantly governed by variations in the coolants' thermophysical properties. Among fluorocarbon coolants, SF10 demonstrates the highest FOM, exceeding Novec 7500 and



(a) Experiment about the server (b) CFD model validation for the server

Fig. 4. Model validation for CFD model.



(a) Experimental system (b) TRNSYS model validation for the system

Fig. 5. Model validation for TRNSYS model.

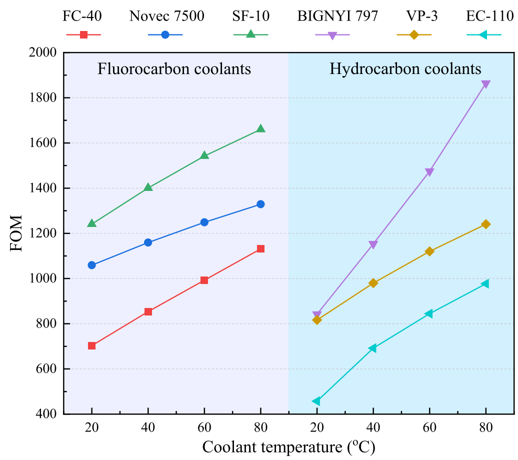


Fig. 6. FOM comparison of different coolants.

FC-40 by 20.8 % and 64.1 % respectively at 40 °C of coolant temperature. This performance hierarchy directly correlates with SF10's optimal combination of thermophysical characteristics: the lowest dynamic viscosity, the elevated heat capacity, and highest thermal conductivity. A similar pattern is observed with hydrocarbon coolants, where BINGYI 797 has the highest FOM, followed by VP-3 and EC-110. When the coolant temperature is set at 40 °C [33], SF10 still leads in terms of FOM, followed by Novec 7500, BINGYI 797, VP-3, FC-40, and EC-110. The superior performance of SF10 and Novec 7500 can be attributed to their low viscosity and high density. It should be note that, other parameter also affects the thermal performance. For instance, despite BINGYI 797 having the highest viscosity at 40 °C, its substantial heat capacity and thermal conductivity contribute to a high FOM. The difference in FOM between SF10 and BINGYI 797 is only 246.9 at 40 °C of coolant temperature. The result highlights the need for a balanced consideration of all thermophysical properties when selecting suitable coolants for SPIC system applications.

Fig. 7 illustrates the relationship between average GPU temperature and coolant flow rates and the results come from the CFD simulation. It can be found that, increasing the flow rate of the coolant significantly reduces the average GPU temperature. This reduction is caused by the enhanced forced flow associated with higher flow rates. For instance, using SF10 as an example, the average GPU temperature drops from 82.6 °C to 66.2 °C as the flow rate increases from 1 m^3/h to 7 m^3/h. The

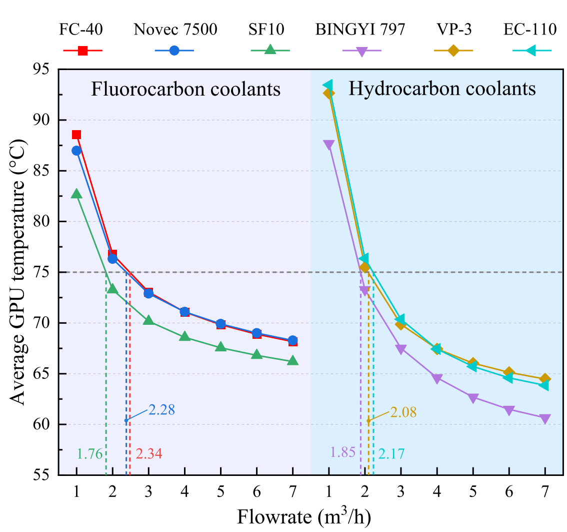


Fig. 7. Average GPU temperature comparison of different coolants.

extent of temperature reduction varies within the flow rate range of 1–7 m^3/h. For fluorocarbon coolants, the temperature decrease is approximately 20 °C, whereas for hydrocarbon coolants, it can be as substantial as 29.6 °C. This is primarily due to significant changes in thermophysical properties. Among fluorocarbon coolants, SF10 results in the lowest average GPU temperature, followed by Novec 7500 and FC-40, when comparing the same flow rates. This trend aligns with the variation observed in the FOM. A similar pattern is observed for hydrocarbon coolants, further underscoring the influence of thermophysical properties on cooling performance. To conduct a fair comparison of different coolants, it is essential to maintain the GPU temperature at a same temperature, i.e., 75 °C [32]. Under this case, the required flow rates for each coolant are specified as follows for a single server, which are 2.34 m^3/h, 2.28 m^3/h, 1.76 m^3/h, 1.85 m^3/h, 2.08 m^3/h, and 2.17 m^3/h for FC-40, Novec 7500, SF10, BINGYI 797, VP-3, and EC-110, respectively, as shown in Fig. 7.

3.2. *Comparison of energy efficiency*

The analysis of annual operating power consumption in Beijing reveals that the SPIC system exhibits lower energy consumption during the spring, autumn, and winter, as shown in Fig. 8. This reduction is primarily attributed to the decreased energy demands of the cooling water pump and the dry cooler during these seasons. However, in the summer, the energy consumption increases significantly due to the higher ambient temperatures. Although the overall trends in operating power with different coolants are similar throughout the year, the actual energy consumption varies notably among them. For example, a substantial 6.8 kW differential in operating power between FC-40 and BINGYI 797 can be found. This pronounced disparity stems primarily from fundamental differences in coolant flow characteristics, particularly in the required flow rate and coolant density, ultimately manifesting as significant system-level power consumption differences.

Fig. 9(a) provides the energy consumption comparison among different coolants, demonstrating that fluorocarbon coolants result in much higher energy consumption than hydrocarbon coolants. Specifically, FC-40 exhibits the highest energy consumption at 109 MW·h, while BINGYI 797 shows the lowest at 49 MW•h. Despite SF10 having the best thermal performance as indicated in Figs. 6 and 7, its high density, nearly double that of BINGYI 797, results in significant energy consumption by the coolant pump. It's important to note that the energy consumption of both the dry cooler and the cooling water pump remains consistent across different coolants, suggesting that the amount of heat transferred is equivalent. However, the substantial energy demand of the coolant pump contributes to variations in the PUE, as shown in Fig. 9 (b). The use of fluorocarbon coolants results in higher PUEs compared to hydrocarbon coolants. FC-40 has the highest PUE at 1.1, which, despite being higher than hydrocarbon coolants, remains significantly lower than typical air-based cooling DC (1.49) [55]. Conversely, BINGYI 797 achieves the lowest PUE, at just 1.046.

3.3. *Comparison of economic evaluation*

Fig. 10 illustrates the TCO of different coolants over the lifetime of SPIC system, highlighting the influence of coolant price and consumption. The result clearly shows that the TCO for fluorocarbon coolants is significantly higher than for hydrocarbon coolants, primarily due to their elevated price. For instance, the TCO for BINGYI 797 is 6.11 million CNY lower than that for Novec 7500. BINGYI 797 stands out with the lowest TCO, attributed to its low density and minimal coolant consumption. While BINGYI 797 carries a higher unit cost than VP-3, its superior density reduces the total coolant mass requirement by approximately 14.9 % in SPIC system applications. This density advantage partially offsets the price differential when considering total fill costs.

Fig. 11 compares the LCC by using different coolants. It can be found

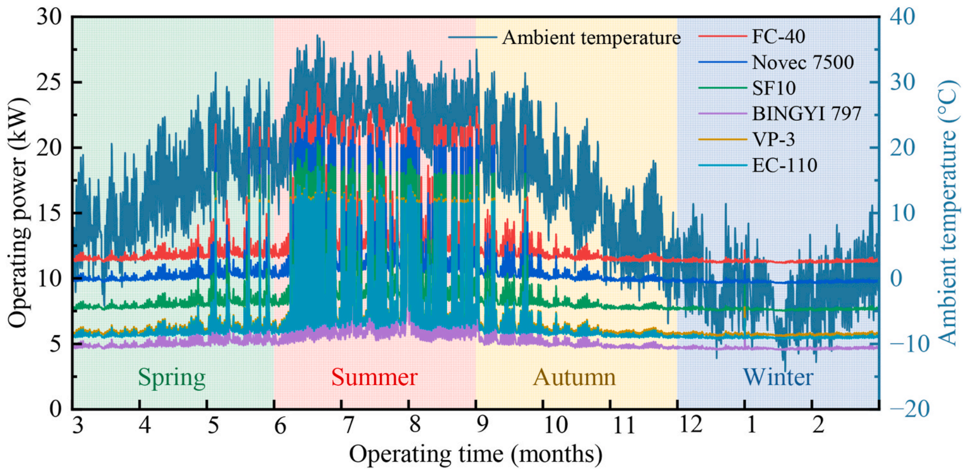
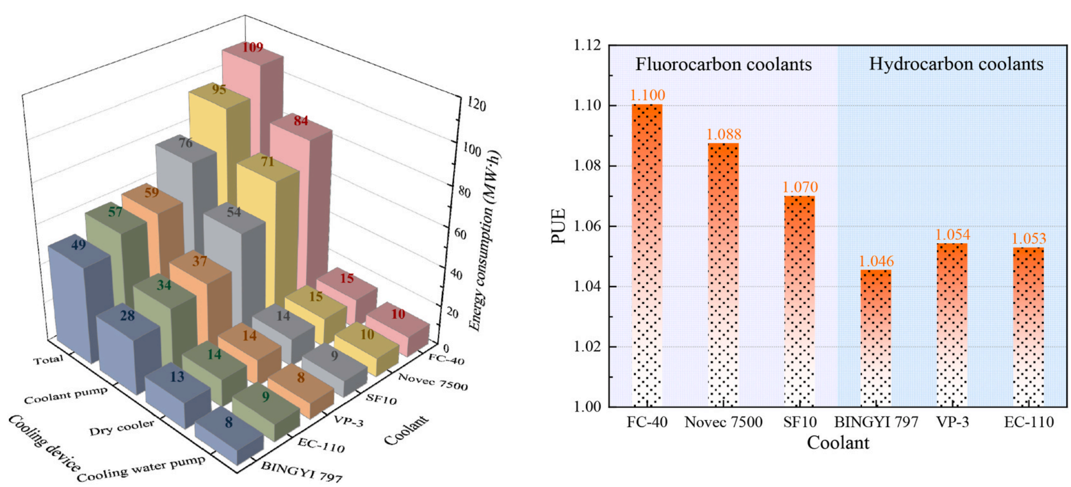


Fig. 8. Operating power by using different coolants.



(a) Annual energy consumption comparison (b) PUE comparison for different coolants

Fig. 9. Energy efficiency comparison by using different coolants.

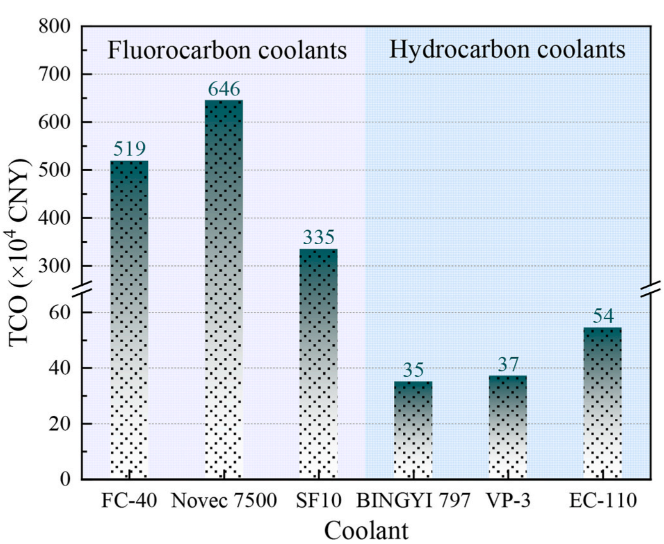


Fig. 10. TCO comparison by using different coolants.

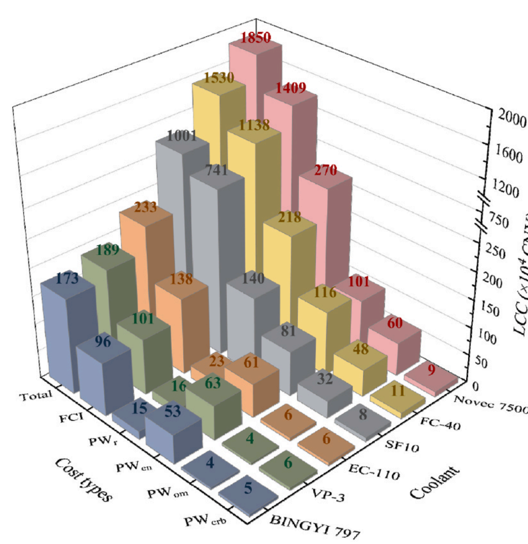


Fig. 11. LCC comparison by using different coolants.

that fluorocarbon coolants exhibit higher LCC than hydrocarbon coolants, which comes from the coolant cost, energy consumption, and operation and maintenance. Among these coolants, Novec 7500 exhibits the highest LCC, whereas BINGYI 797 has the lowest. The use of hydrocarbon coolants can result in substantial savings, reducing LCC by as much as 76.7–90.7 % compared to fluorocarbon coolants. Despite BINGYI 797 having a higher unit price than VP-3, its use leads to a lower PUE, thereby achieving the smallest LCC. For fluorocarbon coolants, the primary contributors to LCC are the coolant purchasing and replenishing costs. In contrast, for hydrocarbon coolants like BINGYI 797, the coolant purchasing cost and energy consumption are the main factors influencing the overall cost.

3.4. Comparison of environmental impact

Fig. 12 illustrates the lifecycle CO_2 emissions associated with using different coolants, highlighting a clear distinction between fluorocarbon and hydrocarbon coolants. Fluorocarbon coolants exhibit significantly higher CO_2 emissions, with FC-40 having the highest emissions primarily due to its high GWP value and energy consumption. For instance, direct CO_2 emissions constitute 94.5 % of the total for FC-40, 25.5 % for Novec 7500, and only 1.03 % for SF10. Notably, the CO_2 emissions from FC-40 are 40 times greater than those from BINGYI 797. Among the hydrocarbon coolants, BINGYI 797 demonstrates the lowest CO_2 emissions, attributed primarily to its low energy consumption and PUE.

3.5. Comparison of different regions

In this section, five cities are selected from various climate zones with distinct ambient temperatures. Moreover, the model also accounts for variations in local electricity prices, carbon prices, and carbon emission factors. The detailed parameters can be found in Table 4. Based on these, the impact of different coolants could be compared.

Fig. 13 presents the energy consumption across different regions when using different coolants. Consistently across all regions, FC-40 exhibits the highest energy consumption and PUE, whereas BINGYI 797 shows the lowest one, aligning with the trends observed in Fig. 9. The performance in different regions varies significantly, with energy consumption differences ranging from 18.9 % to 57.3 % and PUE differences from 1.039 to 1.150. For instance, in Riyadh, the high ambient temperature results in energy consumption and PUE reaching 100 MW·h and 1.092, respectively, when BINGYI 797 is used; while, in Moscow, these values decrease to 43 MW·h and 1.039.

Fig. 14 illustrates the LCC of the SPIC system using different coolants

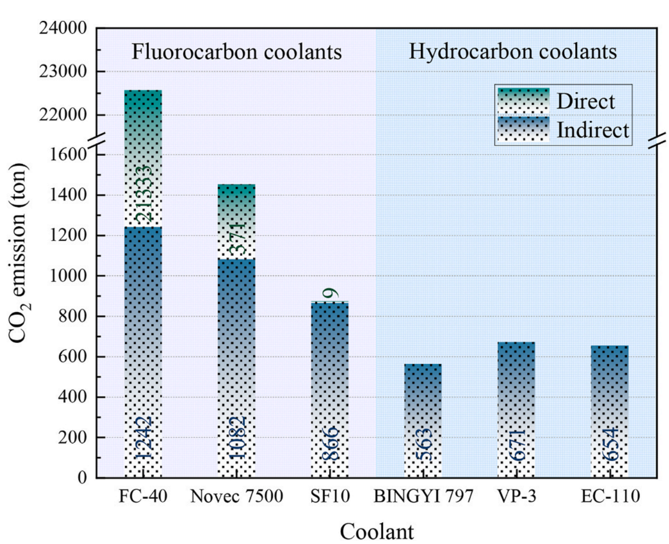


Fig. 12. CO_2 emissions in the lifecycle by using different coolants.

Table 4
Detailed information about 5 regions [56–60].

Regions	Ambient temperature, °C			Electricity price, CNY/kW·h	Carbon price, CNY/ton	Carbon emission factor, $kgCO_2$/kW·h
	Max	Ave.	Min			
Beijing	37.0	12.7	−14.0	0.634	104	0.569
Riyadh	45.5	26.2	4.0	0.487	0	0.506
Moscow	30.5	5.5	−25.0	0.56	0	0.322
Singapore	33.4	27.5	21.5	2.347	133.96	0.412
Los Angeles	33.9	16.7	4.4	1.056	281.73	0.207

across various regions. It is evident that fluorocarbon coolants result in a significantly higher LCC compared to hydrocarbon coolants, irrespective of the region. The variation in LCCs across regions ranges from 13.7 % to 61.5 %, primarily influenced by differences in energy consumption, electricity prices, and carbon pricing. For instance, Singapore exhibits the highest LCC due to its high electricity and carbon prices, as detailed in Table 4. The LCC in Singapore is 17.6–61.5 % higher than that in Moscow. On the other hand, despite Riyadh having the highest energy consumption due to its elevated ambient temperatures, its LCC is 15.9–52.2 % lower than Singapore's, which is attributed to Riyadh's low electricity costs and zero carbon pricing.

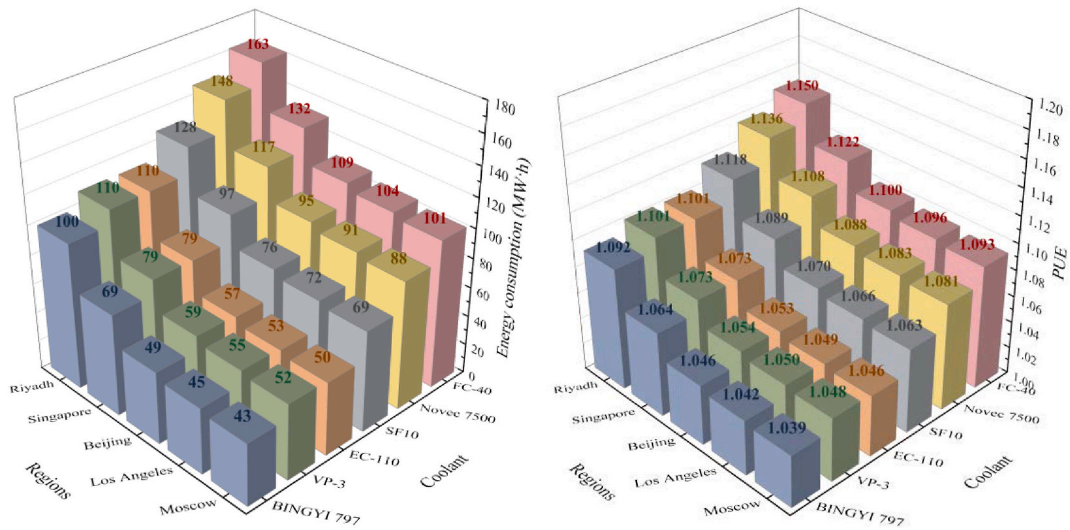
Fig. 15 presents the CO_2 emissions of the SPIC system across various regions, indicating a significant variation ranging from 1.8 % to 81.5 %. The GWP of the coolant plays a crucial role in determining CO_2 emissions. For instance, the use of FC-40 results in relatively consistent CO_2 emissions across regions, varying only between 1.8 % and 5.3 %. Beyond the influence of coolant GWP, CO_2 emissions differ substantially across regions, primarily due to variations in energy consumption and carbon emission factors. For example, Riyadh exhibits the highest CO_2 emissions, while Los Angeles has the lowest, where the disparity in emissions can be as vast as 5.3–81.5 % by using different coolants.

3.6. Comprehensive evaluation

Fig. 16 provides a comprehensive comparison of different coolants, highlighting the trade-offs between thermal performance and other factors such as energy efficiency, economic evaluation, and environmental impact. SF10 offer superior thermal performance due to their low dynamic viscosity, which results in lower average GPU temperatures and a FOM. However, for the practical application of the fluorocarbon coolants, the energy efficiency, economic evaluation, and environmental impact should be fully considered. On the other hand, BINGYI 797 possess characteristics such as low density, high heat capacity, high thermal conductivity, and affordability, which make it more competitive in terms of energy efficiency, economic evaluation, and environmental impact. Among the evaluated six coolants, SF10 and BINGYI 797 are the most promising candidates for fluorocarbon and hydrocarbon coolants, respectively. The findings also underscore the need for the development of new coolants that combine the desirable properties of both types. The ideal coolant for future data center cooling applications would have low dynamic viscosity, low density, high heat capacity, high thermal conductivity, low ODP and GWP, and a low price, balancing thermal performance with economic and environmental considerations.

3.7. Discussion

This study presents a comprehensive evaluation of six different coolants for data center, analyzing their thermal performance, energy efficiency, economic viability, environmental impact, and regional applicability. While SF10 (fluorocarbon coolant) and BINGYI 797 (hydrocarbon coolant) emerge as the most balanced candidates based on



(a) Annual energy consumption

(b) PUE comparison

Fig. 13. Energy efficiency comparison in different regions.

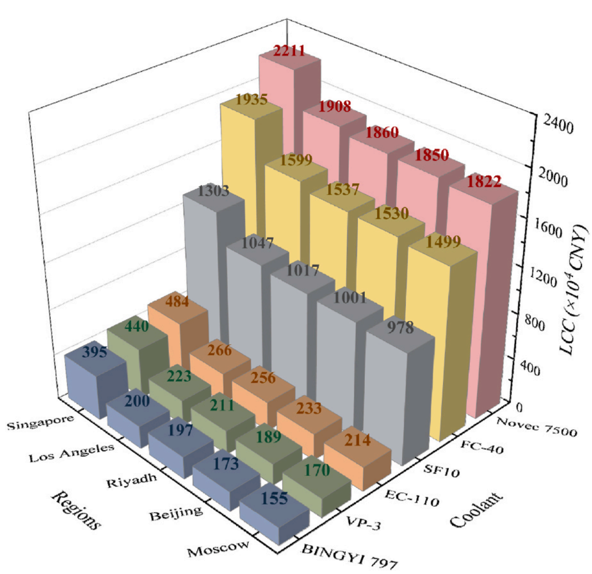


Fig. 14. LCC comparison in different regions.

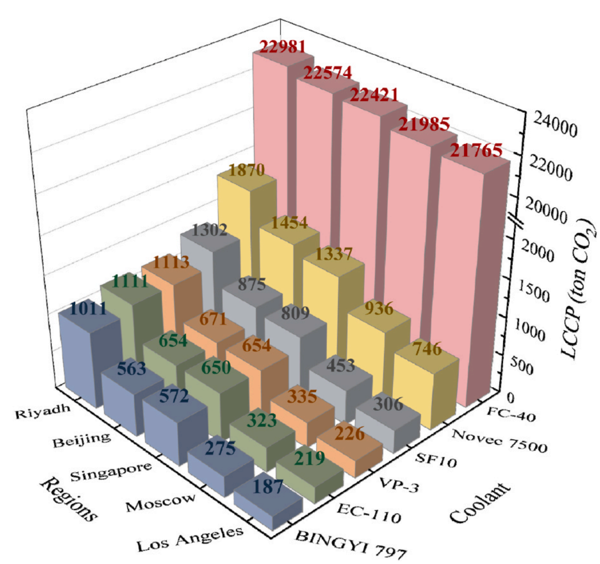


Fig. 15. Carbon emission comparison in different regions.

current metrics, several promising research directions warrant further investigation: (1) Expansion of coolant options: Systematic evaluation of advanced coolant, particularly nanofluids with engineered thermophysical properties, could reveal additional performance benefits. (2) Next-generation coolant development: Targeted modification of key properties (e.g., viscosity, thermal conductivity) through molecular engineering or additive technologies may yield superior coolant formulations. (3) System-level optimization: Implementation studies focusing on energy consumption minimization should be conducted for the recommended coolants, including the pump optimization, heat exchanger redesign, and control strategy development. Moreover, the fluctuation of the IT power should be considered, which would affect the cooling demand. Even at the fixed workload, the IT power may be also different. Combing the CFD and TRNSYS models would be an effective way to solve such issues.

4. Conclusions

To comprehensively assess the different coolant performances used in single phase immersion cooling (SPIC) system and provide a guidance for the selection and development of coolant, this study establishes the both Computational Fluid Dynamics (CFD) model and TRNSYS model. By using the validated models, the comparisons from perspectives of thermal performance, energy efficiency, economic evaluation, environmental impact, and different regions are conducted by using the fluorocarbon and hydrocarbon coolants. Through the results, it can be concluded as follows.

(1) The fluorocarbon coolants show a better thermal performance, and the figure of merit (FOM) and of SF10 is 17.2–50.6 % higher than other coolants at a coolant temperature of 40 °C.
(2) The hydrocarbon coolants show a better energy efficiency, economic evaluation, environmental impact, and BINGYI 797 could reduce energy consumption by 14–54 %, save coolant costs by

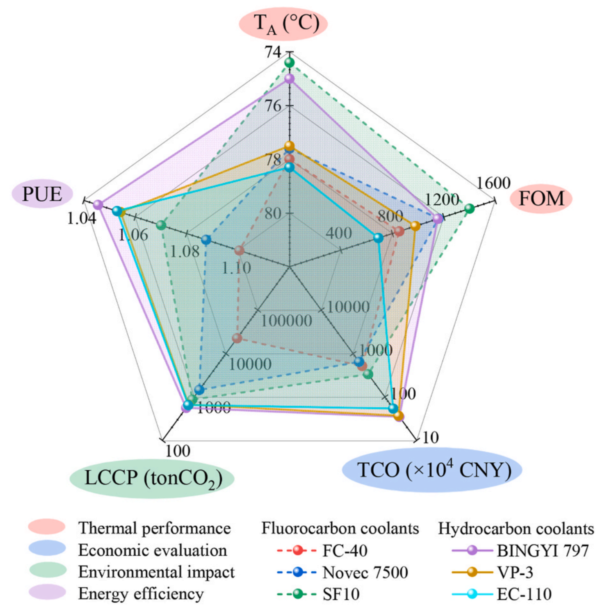


Fig. 16. Comprehensive comparison of different coolants.

5.7–94.6 %, and reduce CO_2 emissions by 13.9–97.4 %, compared to other coolants.

(3) In different regions, the performance is also various. According to different coolants, the energy consumption would range from 18.9 % to 57.3 %, PUE by 1.039–1.150, LCC by 13.7–61.5 %, and CO_2 emission by 1.8–81.5 %.

(4) Among the evaluated six coolants, SF10 (fluorocarbon-based) and BINGYI 797 (hydrocarbon-based) emerge as the most technically viable candidates.

CRediT authorship contribution statement

Xueqiang Li: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Shentong Guo:** Writing – review & editing, Investigation, Data curation. **Haiwang Sun:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Shengchun Liu:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Xinyue Wang:** Writing – review & editing. **Lei Wang:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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